Symmetry breaking and double large primes for the 131-bit Certicom curves (ECC2-131, ECCp-131)

Study `research/certicom131_symlp_20261025` · 2026-10-25 · seed 20261025
Module `src/cryptanalysis/certicom131_symlp` · branch `claude/certicom131-symlp-20261020`

1. Question and scope

Does adding the two standard index-calculus accelerators — symmetry breaking in the point decomposition and a double-large-prime relation graph — move the two non-Koblitz 131-bit Certicom challenges closer to Pollard rho? ECC2-131 is a random binary curve over $\mathrm{GF}(2^{131})$ (field polynomial $z^{131}+z^{13}+z^2+z+1$, cofactor 2); ECCp-131 is over a 131-bit prime (cofactor 1). 131 is prime, so there is no Weil descent to a subfield, and ECC2-131 has $a \notin \mathrm{GF}(2)$, so no Frobenius saving. This study measures the two accelerators on the smallest decomposition arity, $m = 2$, and on 61–127-bit same-family calibration curves, and prices them against rho measured with the same field code on the same host. It makes no key-recovery claim: every end-to-end run uses planted known logarithms and a synthetic target, and the published challenge points are only checked to lie in the subgroup.

2. Method

The factor base is the F_2-subspace $V_\ell = \{x : \deg x < \ell\}$ (Diem / Faugère–Perret–Petit–Renault), with a large-prime domain $V_{\ell'}$, $\ell < \ell'$. A relation $R = \pm P_1 \pm P_2$ is a pair of subgroup abscissae with $S_3(x_1, x_2, x_R) = 0$. In the elementary symmetric variable $e_1 = x_1 + x_2$ the binary S_3 linearises to $t^2 + t = e_1^2 + b/x_R^2$; because $V_{\ell'}$ is closed under addition, $x_1, x_2 \in V_{\ell'}$ forces $e_1 \in V_{\ell'}$, so the **symmetric oracle** sweeps e_1 (each unordered pair once, one batched inversion) against a **baseline** that sweeps x_1 and lifts each point with its own inversion. For ECCp-131 the interval $[0, B)$ is not closed under addition, so e_1 ranges over $[0, 2B)$ and each step costs two square roots; the prime oracle is included to show the saving is special to the binary subspace.

Partial relations (1 or 2 large primes) enter a union–find graph over $\mathbb{Z}/r$ for the 130-bit subgroup order (Gaudry–Thomé–Thériault–Diem); every emitted full relation carries its input ids and is replayed from them. All code is in `src/cryptanalysis/certicom131_symlp` (`gf2w`, `curves`, `sym2`, `sym2p`, `relgraph`, `planted`), tested by `cargo test --lib certicom131_symlp` (18 tests). The driver is `examples/certicom131_symlp_measure`; records are in `runs/`.

3. Measured values

All rates are single-thread on this host, seed 20261025. The IC decomposition step and the rho adding-walk step use the same field implementation in the same process.

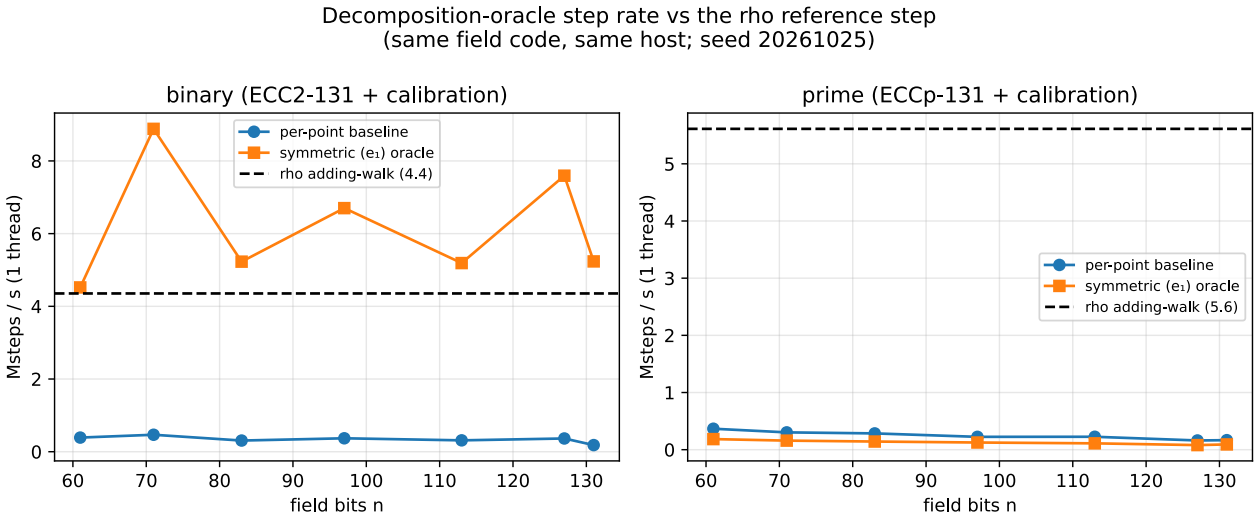


Figure 1: Decomposition-oracle step rate against the rho reference (dashed). Binary: the symmetric e_1 oracle runs above the rho step and 17–29× the per-point baseline, because the baseline pays one field inversion per point and the symmetric sweep batches them. Prime: the symmetric oracle is 2× **slower** than its baseline (two square roots per step), and both are well under rho.

family	curve	n	base M/s	sym M/s	sym/base	IC/rho step
binary	ecc2-131	131	0.18	5.24	28.7×	1.20
binary	cal	127	0.36	7.59	20.8×	1.74
binary	cal	61	0.39	4.52	11.6×	1.04
prime	eccp-131	131	0.17	0.09	0.6×	0.02
prime	cal	61	0.37	0.19	0.5×	0.03

Full table: `tables/measured.md`.

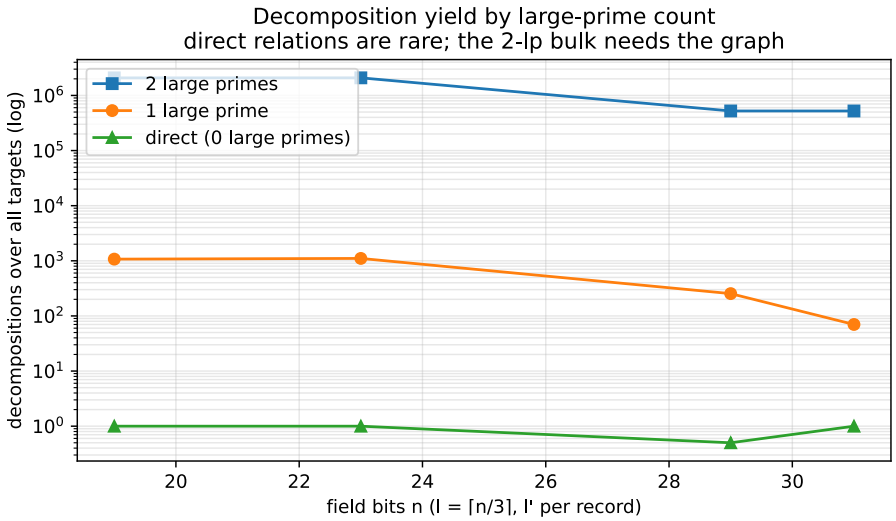


Figure 2: Decomposition yield by large-prime count, full $V_{\ell'}$ sweep over random targets on binary calibration curves. Direct relations (both abscissae in V_ℓ, zero large primes) are a fraction $\sim 5 \times 10^{-7}$ of decompositions; the 2-large-prime relations are the bulk, so without the graph almost every decomposition is discarded.

The planted end-to-end pipeline (`planted.rs`) runs on the **real** 131-bit groups: it closes graph cycles, re-checks every full relation as a point equation in the group, solves the linear system modulo the 130-bit r, and recovers the planted logarithm (both curves, in the test suite).

4. Derived cost model

Every row here is computed, not measured.

At the balancing dimension $d = (n+1)/3$ an $m = 2$ solve needs $\sim 2^{d-1}$ relations at success probability $\sim 2^{2d-n-2}$, i.e. $\sim 2^{2n/3}$ decomposition steps — the method's floor. Rho needs $\sim \sqrt{\pi r/4}$ steps.

curve	n	$\log_2$ steps(IC m=2)	$\log_2$ steps(rho)	$\log_2$ distance
ECC2-131	131	87.3	64.8	22.5
ECCp-131	131	87.3	65.3	22.0

The $m = 2$ exponent $2n/3$ agrees with the merged algebraic ladder's measured trial slope $0.647 \in [0.639, 0.655]$ (PR 2493): enumeration and the linear solve share the $2^{2n/3}$ optimum. Symmetry breaking (a measured 29× per-step saving on binary) and the double-large-prime graph (which rescues the 2-lp bulk of the yield) are constant-factor gains over that baseline. Neither changes the $2n/3$ vs $n/2$ exponent gap, so the distance to rho at $n = 131$ stays $\sim 2^{n/6} \approx 2^{22}$ steps, **before** the linear-algebra floor, which the program gates (`docs/ic/CERTICOM131_IC_PROGRAM.md` §2) place above rho for $m \leq 3$ on both curves.

5. Findings and limits

- **Measured, binary:** the symmetric e_1 oracle's step costs about the same as a rho step on this host (IC/rho 1.0–1.7×) and 17–29× the per-point baseline. This is a genuine, measured improvement to the index-calculus step cost.
- **Measured, prime:** no symmetry saving; the symmetric step is 2× dearer, confirming the subspace-vs-interval asymmetry.

- **Measured yield:** direct relations are $\sim 5 \times 10^{-7}$ of decompositions; the double-large-prime graph is what makes the bulk usable, and the full-size planted pipeline recovers a logarithm through it on both real groups.
- **Distance remaining (derived):** $\sim 2^{22}$ steps at $n = 131$ for $m = 2$, unchanged in exponent by either accelerator; the realistic distance including linear algebra is larger (program gates).
- **Limits:** calibration curves are not point-counted (yield uses random targets, not full solves); the cost model is a model, labelled throughout; no measurement bears on the published challenge points. The accelerators matter for $m \geq 4$, which this study does not reach.